

Sarah Wielgosz Advising Agreement

Term: Fall 2024

Summary

Entering third year in the PhD program. Completed qualifying exam and will be taking the comprehensive exam this year. Has completed most classes (one ML course remaining). Wrapping up past projects and papers from X-lab and undergraduate lab involvement. Prior work includes optimal sensor placement for monitoring generalized targets (ARO-funded), culminating in two conference presentations (October: DC and Japan) and one journal paper.

With transition into the MLDS lab, anticipates a hard pivot into astrodynamics research for the remainder of the PhD. Past research is unlikely to fuse into the dissertation topic. This requires expedited publishing timelines to remain on track for graduation ($\leq 5-6$ years). This remains achievable but requires consistent, dedicated effort (7–8/10).

Priority is to complete a minimum viable research product by the end of the semester, with the goal of presenting at the Astrodynamics Specialist Conference (ASC) in Summer 2025 and transitioning rapidly to a journal paper.

The current project focus is fault detection using Neural Ordinary Differential Equations (Neural ODEs). This requires proficiency in scientific machine learning and representative simulators of dynamical systems with perturbations. Initial systems include a perturbed two-body problem (thrusters, J2). A higher-fidelity system will likely require Basilisk. By semester end, at least one Neural ODE-based fault detection result is expected.

Funding includes an MRC fellowship (AY 2024–25) and NSF-GRFP (AY 2025–28), allowing research flexibility. The goal is to identify and commit to a compelling dissertation direction this semester, with Neural Lambert solvers as a backup.

Primary Project: Fault Detection using Neural ODEs (Secondary: John)

Secondary Project: Neural Lambert Solvers (Primary: Kruti)

Official Deliverables

- Literature review on fault detection in astrodynamics and Neural ODEs — **Completed**
- Minimum viable fault detection algorithm for spacecraft in orbit — **IP**
- Comprehensive exam — **Incomplete (scheduled)**
- Concrete program schedule — **IP**
- Consistent work schedule — **½ completed**

Research Goals

- MAE class tied to research — **Completed**
- Organize and complete comprehensive exam — **½ completed**
- Establish boundaries for other projects — **Completed**
- Proficiency in Git, Python, Neural Networks — **Completed (significant progress)**
- Conference abstract for ASC 2025 — **Confused**
- Read three ASC 2024 papers — **Partial**
- Attend two conferences (October) — **Completed**
- Networking events — **Completed**
- (Stretch) Basilisk setup with multiple flight modes — **Incomplete**

Literature Reviews (½ Progress)

Topics

- Fault detection in spacecraft
- Neural ODEs for fault detection

Papers

- SALAMANDER Neural ODEs — Jordan Murphy

Textbooks

- Selected chapters from *Dive Into Deep Learning*

Minimum Viable Product

Environment

- Onboarding Exercise #2 — **Completed**
- Neural ODE for orbit problem — **Completed**
- Injected orbit fault — **Completed**

Solution

- Identification of single faults — **IP**
- Identification of continuous faults / perturbations — **IP**

Evaluation

Category	Score	Notes
Research	Average	Also working with Dr. Xu's lab
Learning	Average	
Writing	Good	Writing, but not leading journal paper
Verbal	Good	
Presentation	Average	
Academics	Excellent	
Time Management	Fair	
Networking	Good	
Volunteering	NA	
Work Life Balance	Good	

Term: Spring 2025

Summary

Halfway through the third PhD year and six months in the MLDS lab. Focused on neural networks and machine learning fundamentals, Neural ODE literature review, final coursework, and concluding X-lab projects. Conflicting priorities and amorphous outcomes hindered focus; the goal moving forward is streamlined research processes and communication.

Priority shifts to publishing and defending, producing conference-ready deliverables at AAS/AIAA venues. Initial results will demonstrate the utility of Neural ODEs in astrodynamics.

Proposed Papers

1. **Characterization of Neural ODEs for Astrodynamics**
 - Known dynamics (2BP, CR3BP)
 - Unknown dynamics (thrust, hi-fi gravity)
 - Stochastic systems (telemetry, noise sensitivity)
2. **Linear Representation of CR3BP in Latent Coordinates**
 - Latent dimensionality manipulation
 - Predictive accuracy in latent space
 - Recovery of semi-stable orbital elements
 - Uncertainty propagation
3. **Online Construction of Latent Dynamics via Neural ODEs**
 - Filtering with DMC
 - Transfer learning
4. **Fault Detection and Diagnostics**
 - Latent-space covariance-based detection
 - Online model adaptation
 - Control for fault correction

For ASC, the focus is Paper 1 (known and unknown dynamics). Full draft by April; stochastic results ready by conference.

Primary Project: Fault Detection using Neural ODEs

Secondary Project: Neural Lambert Solvers

Official Deliverables

- Complete comprehensive exam
- Migrate code to MLDS org with updated GitHub workflow (≥ 10 PRs)
- Full conference paper draft by April
- Curiosity experiment using all 192 cores

Research Goals

- Integrate Weights & Biases
- Read ≥ 5 SFM papers
- Maintain public research log
- Advanced Git proficiency (interactive rebasing)
- Learn Basilisk and document MLDS fork
- Interface with undergraduates

Evaluation

Category	Score	Notes
Research	Average	
Learning	Average	
Writing	Good	
Verbal	Good	
Presentation	Fair	
Academics	NA	
Time Management	Average	
Networking	Excellent	
Volunteering	Good	
Work Life Balance	Good	

Term: Summer 2025

Summary

Concluding third PhD year. ASC abstract accepted; comprehensive exam completed. Continued Neural ODE development for astrodynamics dynamics modeling, focusing on training intuition and representation learning.

Recent Accomplishments

- Comprehensive exam
- ASC abstract accepted
- Improved Git and PR fluency
- Strong semester start
- Positive lab culture contributions
- Research notes maintained

Areas of Improvement

- More consistent schedule and output
- Use research notes for rubber-duck debugging
- Optimize for future self

Official Deliverables

- Full conference paper
- ASC presentation
- Preliminary stochastic results
- Increased CR3BP fluency

Research Goals

- Neural ODEs v1.0 release
- Journal paper first draft
- ≥ 1 figure + PR per advising meeting
- Integration with MLDS-ML
- Project page for conference paper
- Ray-based parallelization
- Zaratan / SLURM experiments
- Continued research log

Evaluation

Category	Score	Notes
Research	Fair	Pace needs to increase
Learning	Average	Conference exposure highlighted gaps
Writing	Average	Adequate under constraints
Verbal	Good	Clear 1:1 updates
Presentation	Good	ASC presentation solid
Academics	NA	

Category	Score	Notes
Time Management	Poor	Missed deadline
Networking	Average	Self-assessed
Volunteering	NA	
Work Life Balance	Average	

Term: Fall 2025

Summary

Entering fourth PhD year. Submitted and presented ASC paper on Neural ODEs for planar 2BP and CR3BP. Identified segmentation as the key performance mechanism, enabling faster progress going forward.

Time management failures led to missed deadlines and lost opportunities. Future abstracts require near-complete paper material beforehand.

Recent Accomplishments

- Conference paper and presentation
- Professional-quality slides
- Completion under pressure

Areas of Improvement

- Time management realism
- Debugging via simplification
- Conservative communication of timelines

Official Deliverables

- Mature Neural ODE work — **WIP**
- Journal paper — **Incomplete**
- Identify ASC 2026 ideas — **Completed**
- ASTRA showcase poster — **Completed**

Research Goals

- Neural ODEs v1.0 — **WIP**
- MLDS-Astro v0.1 — **WIP**
- ≥ 1 figure per advising meeting — **Completed**
- Resume research notes — **Completed**

Evaluation

Category	Score	Notes
Research	Fair	
Learning	Good	Strong NN + astrodynamics growth
Writing	Fair	
Verbal	Good	
Presentation	Good	Two best-figure awards
Academics	NA	
Time Management	Fair	Missed deadline
Networking	Good	
Volunteering	Good	Mentorship
Work Life Balance	Good	

Term: Spring 2026

Fourth-year PhD student continuing research on the use of neural ODEs to learn latent models of nonlinear dynamical systems. The primary priority for the Fall semester was to reanalyze the prior ASC conference work with more comprehensive metrics, ablation studies, and visualizations. This deeper evaluation revealed performance behaviors that were inconsistent with prior intuition, making earlier conclusions—such as the value of certain feature layers and segmentation strategies—less definitive.

While overall research velocity was slower than desired, there were meaningful signs of progress. Intermediate visualization capabilities have matured significantly and were recognized with multiple best figure awards. In addition, debugging practices have improved, and daily research updates toward the end of the semester were consistently high quality and demonstrated strong research discipline and professional habits.

Moving forward, the central priority remains a rigorous recharacterization of the neural ODE approach from the ASC work, with the goal of identifying modeling strategies that produce consistent performance in data-sparse regimes. Establishing a clear path to a journal submission is essential. Viable avenues for novel contribution include application to periodic orbit discovery and/or a transition toward latent linear ODE formulations. The value of these approaches should become more apparent when applied to the CR3BP, particularly if they enable accurate modeling of long-horizon periodic, resonant, or quasi-periodic trajectories.

As part of the Astrodynamics Program, an additional expectation for the semester is the development of an internal astrodynamics library capable of generating advanced CR3BP trajectories suitable for training and evaluating machine learning models.

Areas of Growth / Improvement

Research Closure and Deliverable Focus

The primary area for improvement is consistent delivery of formal research outputs. Progress should be structured around producing concrete artifacts—such as a manuscript, technical report, or documented null result—by the end of the semester. Even when results are inconclusive, a written deliverable provides evidence of progress, consolidates experiments and visualizations, and creates a foundation that can be extended when stronger results emerge.

Research Narrative Development

There is an opportunity to place greater emphasis on identifying and shaping a clear research story earlier in the process. Defining the central question, expected contribution, and evaluation criteria up front can help guide experimentation and reduce exploratory drift.

Engagement and Group Contribution

More active participation in group meetings is encouraged, including asking questions when concepts are unclear and contributing to technical discussion. This supports both individual understanding and overall group engagement.

Accountability and Communication Cadence

Maintaining regular progress visibility through brief updates (e.g., Slack check-ins or milestone summaries) can help sustain momentum and enable earlier course correction when needed.

Official Deliverables

- Submit journal paper demonstrating a validated contribution (e.g., application of Neural ODEs to periodic orbit discovery or an alternative well-supported research outcome)

- Deliver a minimum viable implementation of latent linear ODE models
- Develop a shared astrodynamics + ML internal toolset for CR3BP trajectory generation (Astrodynamics Program)

Research Goals

- Prepare ASC paper for Breakwell Award consideration
- Re-establish consistent Git workflow practices (branching, pull requests, reproducible history)
- Review Ethan Burnett's work on relative motion modal decomposition
- Review the theses of Gavin Brown and Luke Peterson